Part H: Bias–Variance & Model Diagnostics

Task 26: Bias–Variance Analysis

- **Simple Linear Regression (SLR):**
  - **Bias:** High, because the model is too simple (only one predictor). It cannot capture all factors affecting house price.
  - **Variance:** Low, because the model is stable and not sensitive to small changes in data.
  - **Result:** Underfitting risk — predictions are too simplistic.
- **Multiple Linear Regression (MLR):**
  - **Bias:** Lower than SLR, since more predictors are included. Captures more real-world complexity.
  - **Variance:** Moderate, as adding predictors increases sensitivity but still manageable.
  - **Result:** Balanced model — usually better fit without extreme instability.
- **Polynomial Regression:**
  - **Bias:** Very low, because the model can capture nonlinear relationships.
  - **Variance:** High, especially with higher degrees (degree ≥ 3). Predictions may fluctuate heavily with small data changes.
  - **Result:** Overfitting risk — excellent fit on training data but poor generalization.

Task 27: Effect of Model Complexity on Prediction Error

- As model complexity increases (from SLR → MLR → Polynomial):
  - **Bias decreases** (model explains more variance).
  - **Variance increases** (model becomes more sensitive to noise).
- **Prediction Error (Test Data):**
  - Initially decreases as complexity helps capture patterns.

    - Eventually increases if the model becomes too complex (overfitting).
- This trade-off is known as the **Bias–Variance Trade-off**.

Task 28: Best Balance of Bias and Variance

- **Simple LR:** Too biased → underfits.
- **Polynomial Regression (high degree):** Too much variance → overfits.
- **Multiple LR:** Best balance → captures multiple predictors, reduces bias, and keeps variance at a reasonable level.
- **Conclusion:** Multiple Linear Regression provides the most practical balance between bias and variance for housing price prediction.